SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I THIRU KUMARAN S / 192521149 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thiru Kumaran S

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded :

1. Compare Type - 1 and Type - 2 Hypervisors for a cloud Data Center:

A hypervisor is software that creates and manages virtual machines (VMs), allowing multiple operating systems to run on the same physical computer.

Type - 01 Hypervisor (Bare - Metal)

* Runs directly on physical hardware.
* provides high performance and low overhead.
* offers strong isolation between virtual machines.
* Suitable for large cloud datacenters.
* eg: VMware ESXi, Microsoft Hyper-V.

Type - 02 Hypervisor (Hosted)

* Runs on top of a host operating system.
* Has more performance overhead than Type-1.
* Depends partly on the security of the host OS.
* Easier to install and use for testing & development
* eg: oracle virtualBox, VMware workstation

Justification:

For a cloud datacenter hosting mission - critical applications, type - 1 hypervisor is the better choice because it provides better performance, efficient resource utilization, strong VM isolation.

Conclusion:

Type-1 hypervisors are more suitable for mission critical cloud environments, while type-2 hypervisors are better for personal use, testing, and development.

2. Workload isolation in cloud virtualization:

A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same physical hardware. virtualization helps achieve this by running each workload in a separate virtual machine (VM).

How virtualization Supports Isolation:

1. Separate virtual Machines:
2. Memory Isolation
3. Network Isolation.
4. Access Control.

Two major Attack surfaces:

1. Hypervisor vulnerabilities:

If attackers exploit a hypervisor vulnerability, they may escape a VM and access the host or other VMs.

2. Virtual Network Attacks:

Attackers may exploit weak network configurations to intercept data or access unauthorized workloads.

Security Measures:

* Regularly update and patch the hypervisor.
* use strong authentication and access Controls.
* Configure virtual firewalls and network segmentation.
* Monitor VM activities and detect suspicious behavior.

Conclusion:

Virtualization enables multiple workloads to share the same hardware while maintaining isolation. using secure hypervisor configurations, network segmentation, and regular monitoring helps protect finance, student data, etc.

(3) VM Host utilization Analysis & Corrective Actions:

A cloud provider observes the following VM host utilization

* CPU utilization: 92%
* Memory utilization: 88%.
* Storage I/O latency: 35ms
* Average VM boot time: 170 sec

1. Analysis of Bottlenecks

CPU Bottleneck: CPU utilization is 92%,

Memory Bottleneck: Memory utilization is 88%.

Storage Bottleneck: Storage I/O latency is 35ms,

VM Boot Time: An avg boot time of 170 sec.

2. Corrective Actions:

* optimize CPU:
* Increase Memory:
* improve storage
* Optimize VM Booting
* Monitor performance:

Conclusion:

The high CPU & memory utilization, together with increased storage latency, indicate resource contention on the VM host. By optimizing resource allocation, upgrading storage, & monitoring performance, the cloud provider can improve VM performance.

4. How Software Defined Datacenter (SDDC) improves Agility.

A Software Defined Datacenter (SDDC) is a datacenter where Computing, storage, & networking resources are managed through Software instead of depending mainly on manual hardware

1. Compute virtualization:
   * creates multiple virtual machines on physical servers.
   * Allows resources to be allocated according to work load requirements.
   * Helps deploy applications quickly.

2. Storage virtualization:
   * Combines storages resources into a single
   * Allows flexible storage allocation.
   * improves storage utilization and simplifies management.

2. Proposed Secure Identity Design:

* Identity Federation:
* Single Sign-on (SSO)
* Standard protocols:
* Role-Based Access Control (RBAC)
* Multi-factor Authentication (MFA).

3. Privacy protection:

* Encrypt sensitive data during storage and transmission.
* Share only the minimum necessary user information.
* Maintain audit logs to track access & detect suspicious activity.
* Apply consistent security policies across all cloud providers.

Conclusion:

Multi-cloud identity mismatches can be solved using identity federation, SSO, standard authentication protocols. and role-based access control.

3. Network virtualization:

Advantages of SDDC:

* Agility
* Scalability
* Automation
* Cost efficiency
* Centralized management;

Conclusion:

SDDC improves datacenter agility through compute, storage, and network virtualization. It provides faster deployment, flexible resource allocation, automation, and better scalability compared with a traditional datacenter.

5 Multi-cloud federation:

Identity & privacy Design:

A multi-cloud federation allows different cloud providers to work together and share resources securely. Identity mismatches occur when different providers use different methods, user identities,

1. Analysis of the problem:

* Different cloud providers may use different identity formats
* user authentication may fail between providers.
* Access permissions may not match across clouds.
* Sensitive data may be exposed if identity verification is weak.